Reinforcement Learning vs. Rule-Based Policies for Dynamic Credit Threshold Control: A Simulation Study

Evgeny Baulin[a,1,*], P. P. Lukianchenko[b,2]

[a] *HSE University, Faculty of Computer Science, Bachelor's Programme "Data Science and Business Analytics", Moscow, Russia*
[b] *Trusted AI Research Center, Ivannikov Institute for System Programming of the RAS (ISP RAS), Moscow, Russia*

Abstract

Portfolio-level credit acceptance thresholds are revised periodically in practice, which makes the decision sequential rather than one-shot. Each week the controller sets two acceptance thresholds, one for new clients and one for repeat clients, and then observes delayed repayment and default outcomes over subsequent weeks. We evaluate six deep reinforcement learning (RL) algorithms (DQN, Double-DQN, PPO, SAC, A3C, and A2C) against five rule-based baselines in a calibrated synthetic credit-portfolio simulator across six market scenarios and four observation-space sizes (12D, 20D, 30D, 50D). Data-driven weekly threshold control is one concrete way intelligent systems can improve business lending decisions, because adjusting thresholds to current portfolio and market signals can reduce realized default rates while protecting profit, and this study identifies when such automation pays

*Corresponding author.
Email addresses: eabaulin@edu.hse.ru (Evgeny Baulin), p.lukyanchenko@ispras.ru (P. P. Lukianchenko)
[1]ORCID: 0009-0004-9030-2893.
[2]ORCID: 0000-0002-1022-6041.

off. The simulator is a calibrated proof-of-concept test bench, not a substitute for real banking data. Only observation dimensionality varies across the four conditions, with the simulator, reward, seeds, and evaluation protocol held fixed. We find that the constraint-aware weekly policy remains the best overall controller by cumulative reward (the primary ranking metric, with realized profit 3,802,547) and is statistically tied with the profit-oriented policy on realized profit, and RL still does not win the reward ranking in any scenario or dimension. Agents are trained for 1,500 episodes per seed, a substantially larger budget than a short pilot run, and under this budget the best RL agent closes almost all of the profit gap to the leading baseline, from -9.4% under a short 50-episode budget to between -0.5% (12D, SAC) and -2.5% (30D, DQN), and even wins raw profit in two of six scenarios at 12D (SAC). The strongest RL agent varies by dimension (SAC at 12D and 50D, A2C at 20D, DQN at 30D), so no single algorithm dominates across the observation sizes. A controlled state-dimensionality comparison on the canonical DQN agent shows that 30D significantly outperforms both 12D and 50D on cumulative reward, and this is not only a reward effect: the best RL agent undercuts both profit-seeking baselines on default rate in five of six scenarios at 30D and 50D, versus one of six at 20D and zero of six at 12D, so the conditional risk-side advantage of richer state emerges specifically once the observation is rich enough. A delayed-evaluation check in the adverse-stress scenario shows near parity between DQN and the static-threshold baseline early in the horizon and a large, highly significant DQN advantage by the horizon midpoint, supporting a "locally tied, globally better" pattern. Episode-static behavior, in which a controller fixes a threshold

pair at the start of an episode and barely adjusts it, is no longer universal: it is specific to the DQN family at 12–30D, while SAC and the actor-critic agents adjust thresholds within the episode at every dimension. We conclude that, given a sufficient training budget, RL becomes highly competitive with well-designed rule-based weekly threshold search, though the strongest rule-based policies still win the primary reward ranking in every scenario and dimension tested.
Keywords: Reinforcement Learning, Credit Scoring, Threshold Control, Markov Decision Process, Delayed Rewards, Portfolio Simulation, Rule-Based Policies, State Dimensionality

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1. Introduction

A credit manager who holds a single static acceptance threshold loses money as the portfolio composition, the capital usage, and the market conditions drift between decisions. In practice, banks and consumer-lending institutions revise acceptance thresholds periodically rather than once. The fraction of applicants to approve this week depends on the current portfolio composition, outstanding capital commitments, rolling default rates, and market conditions that evolve between decisions. The weekly acceptance decision is therefore a recurring control problem, not a one-shot cut-off. Much of the academic literature has instead treated threshold selection as a one-shot classification problem, estimating a default probability, choosing a cut-off, and evaluating by AUC or Gini, which ignores the sequential nature of actual threshold management (Thomas, 2017; Lessmann et al., 2015). That gap between static cut-off optimisation and dynamic portfolio management

motivates the present study.

We build and evaluate an intelligent decision-support module for weekly threshold control, and we compare learned (RL) controllers against strong rule-based expert policies. Data-driven weekly threshold control is one concrete way artificial intelligence can improve business lending decisions. By adjusting acceptance thresholds to current portfolio and market signals, it can reduce realized defaults while protecting profit, and the present study identifies the conditions under which such automation pays off. Reinforcement learning provides a natural framework for decisions with delayed consequences (Arulkumaran et al., 2017). An RL agent observes the state of the portfolio, selects a pair of weekly thresholds, and receives a reward signal that reflects realized portfolio cash flows rather than immediate cohort expectations. The delayed-outcome setting is central: a loan accepted at week t may not default or repay until week $t + \Delta$, so the correct evaluation object is the full portfolio trajectory, not the immediately observable cohort. Herasymovych et al. (2019) showed that a DQN agent can learn to adjust a single acceptance threshold in a simulated credit environment with delayed outcomes, establishing the structural ideas on which this study builds.

Calibration disclaimer. All experiments in this paper use a synthetic simulator calibrated to published retail lending statistics (Bank of Russia supervisory data and ECB SSM loan loss reports) as a controlled proof-of-concept test bench. Synthetic simulation is standard practice in RL-based credit research because real loan-level panels with weekly cohort granularity and observable repayment and default events are not publicly available (Herasymovych et al., 2019; Lessmann et al., 2015). The simulator provides full

control over market regime, segment composition, and shock timing, which makes the multi-scenario robustness evaluation possible. Results should be interpreted as evidence about algorithmic behaviour under these controlled conditions, not as forecasts for any specific production portfolio.

The paper extends the framework of Herasymovych et al. (2019) with four contributions: (i) split weekly thresholds for new and repeat client segments, (ii) a controlled state-dimensionality experiment comparing 12D, 20D, 30D, and 50D observation sizes under a fixed protocol, (iii) five economically meaningful rule-based baselines as the comparison standard, and (iv) statistical hypothesis tests on paired run-level data. The central research question is:

> *Under which conditions does reinforcement learning provide a robust improvement over rule-based threshold policies in a delayed-outcome credit-scoring environment?*

The data give a conditional answer. Strong rule-based weekly search wins every scenario and dimension by the primary reward ranking, and with a training budget large enough for the RL agents to approach convergence (1,500 episodes per seed) the best RL agent closes almost all of the profit gap to the leading baseline and even wins raw profit in two of six scenarios at 12D. Richer-state RL agents also produce lower portfolio default rates than the profit-seeking baselines in a majority of scenarios once the state is rich enough (five of six scenarios at 30D and 50D, versus one of six at 20D and zero of six at 12D), while the risk-aware weekly policy stays the single lowest-default controller overall. This points to a conditional risk-side advantage that becomes visible once the profit ranking is supplemented by credit-quality metrics, and that itself depends on observation dimensionality.

2. Related Work

Traditional credit scoring estimates default probability at the application level and selects a cut-off that balances interest income against expected loss (Thomas, 2017; Hand & Henley, 1997; Siddiqi, 2006). Benchmarking studies have shown that classifier rankings depend on the dataset and the evaluation protocol, and that logistic regression and gradient-boosted trees are difficult to outperform with more complex classifiers (Lessmann et al., 2015; Baesens et al., 2003). The same conclusion recurs in more recent work: Gunnarsson et al. (2021) find that deep neural networks do not reliably beat gradient boosting on standard credit datasets, and reviews of consumer credit risk assessment reach a similar verdict on where modelling effort actually pays off (Zhang & Yu, 2023). Application-oriented research has therefore shifted from raw classifier accuracy toward feature enhancement and ensembling (Yang et al., 2024) and toward explainability for credit decisions (Nallakaruppan et al., 2024), reflecting the practical priorities of intelligent lending systems.

These advances still treat the accept/reject cut-off as a static one-shot tuning step. A second line of work reframes threshold choice as a cost-sensitive economic decision rather than a classification metric (Baesens et al., 2003), and Bellotti & Crook (2009) note that under Basel II/III capital adequacy requirements the threshold affects outstanding exposure, realized losses, and regulatory capital in ways that make static cut-off selection insufficient once market conditions or portfolio composition drift over time. Once the decision is recognised as recurring and state-dependent, it becomes a sequential control problem, which is the natural domain of reinforcement learning.

Reinforcement learning addresses problems with delayed consequences

through the Markov Decision Process (MDP) framework (Dayan & Niv, 2008; Arulkumaran et al., 2017). The deep RL algorithms used in this study illustrate the main design axes. DQN couples Q-learning with experience replay for off-policy sample efficiency on discrete action spaces, Double-DQN reduces the overestimation bias of vanilla DQN (van Hasselt et al., 2016), PPO is an on-policy policy-gradient method with stable updates (Schulman et al., 2017), SAC is an off-policy maximum-entropy actor-critic (Haarnoja et al., 2018), and A3C and A2C are asynchronous and synchronous actor-critic variants. A broad survey of these methods appears in Arulkumaran et al. (2017). In this study A2C, PPO, and SAC use the Stable-Baselines3 implementations (Raffin et al., 2021), while DQN, Double-DQN, and A3C use matched custom PyTorch implementations.

Applying RL to credit control is more recent and still thin. Herasymovych et al. (2019) optimise the acceptance threshold of a credit-scoring model with RL in a simulated environment with delayed outcomes, and Kiatsupaibul et al. (2024) survey reinforcement learning for credit scoring and underwriting and note how few controlled comparisons against strong rule-based policies exist. The direct precursor (Herasymovych et al., 2019) compared a DQN agent against a single static threshold in one scenario using a shared, non-split threshold. No prior work has simultaneously evaluated split thresholds for distinct borrower segments, a controlled observation-dimensionality comparison, strong economically motivated rule-based baselines, and six market scenarios with statistical testing. That combination is the gap this paper fills, and it motivates the three hypotheses stated in Section 4.1.

3. Problem Formulation and Methodology

3.1. MDP Formulation

We model the weekly threshold-control problem as a discrete-time MDP $(\mathcal{S}, \mathcal{A}, \mathcal{P}, r, \gamma)$. The **state** $s_t \in \mathbb{R}^d$ ($d \in \{12, 20, 30, 50\}$) is a portfolio summary vector observed at the beginning of interactive week t. It includes approval rates, default rates, realized profit and NPV, capital usage, outstanding exposure, segment shares, and, at higher dimensionalities, lagged and rolling history. The **action** is a pair of acceptance thresholds $a_t = (\tau_t^{\text{new}}, \tau_t^{\text{repeat}}) \in \{35, 40, \ldots, 85\}^2$: applications from new (repeat) clients with credit score above τ_t^{new} (τ_t^{repeat}) are accepted. This yields $11 \times 11 = 121$ joint actions per week. The **reward** is described in Section 3.4. The discount factor is $\gamma = 0.98$.

Figure 1 shows the weekly decision loop as a Markov decision process. The figure makes explicit that each weekly action is scored only through delayed cash flows, so the controller must account for outcomes that arrive several weeks after the threshold is set.

3.2. Simulation Environment

Each episode consists of three phases. (1) An **8-week warm-up** prefills the loan book using default thresholds before the controller begins acting. (2) During the **52-week interactive horizon** the controller sets weekly thresholds. The simulator generates applications from Gaussian credit scores clipped to $[0, 100]$ for new and repeat segments, applies a logistic default-probability mapping, accepts loans above the threshold, and schedules repayment, default, and recovery events in a forward event queue. (3) **Terminal**

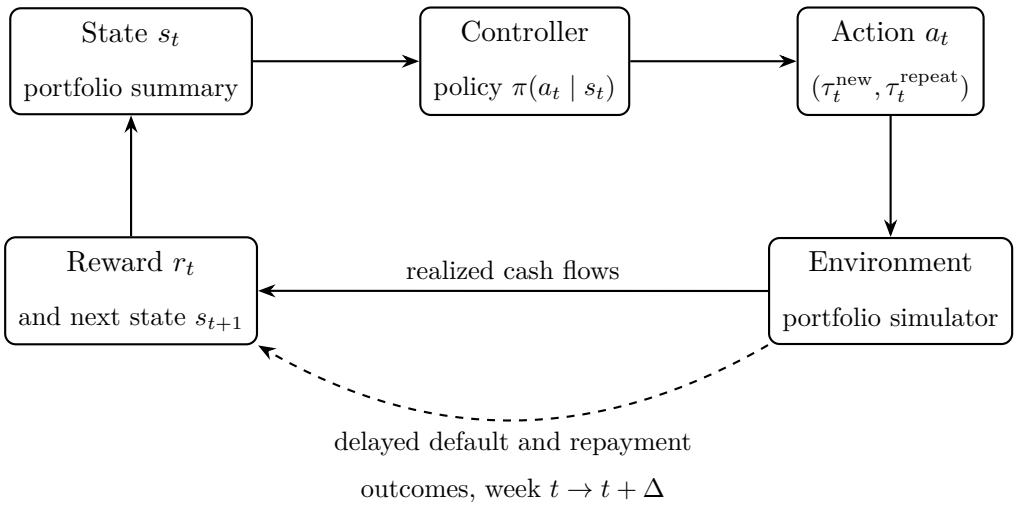


Figure 1: Weekly threshold control as a Markov decision process. The dashed arrow marks the delayed default and repayment outcomes that link an action taken at week t to rewards realized at later weeks $t + \Delta$.

settlement resolves all pending cash flows at episode end so that loans originated near the horizon are not mechanically penalized. Default probabilities are calibrated to target 4–6% in the base scenario and 9–14% under adverse stress, consistent with published retail lending statistics.

Figure 2 shows the three-phase episode structure and the within-week loan lifecycle. The figure shows that outcomes from accepted loans are pushed into a forward event queue, so repayments, defaults, and recoveries feed back into later weeks rather than into the week of origination.

In Table 1, the six evaluation scenarios and their primary stress mechanism are listed. The scenarios probe distinct portfolio-management weaknesses, so a controller that performs well across all of them is robust rather than tuned to a single regime.

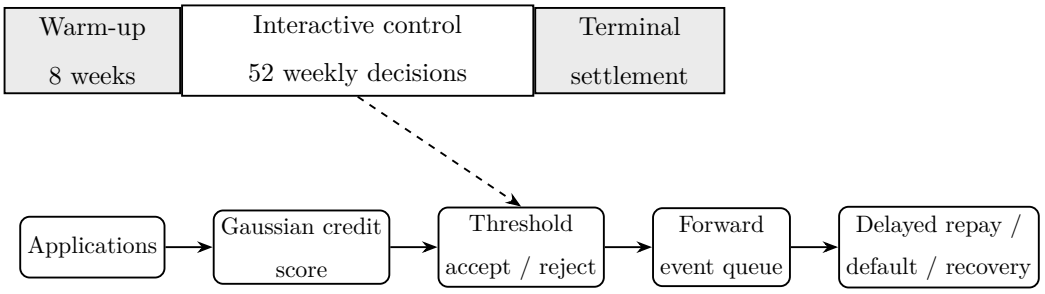


Figure 2: Episode structure and loan lifecycle. The top row is the episode timeline (8-week warm-up, 52-week interactive weekly control, terminal settlement), and the bottom row is the within-week sequence from application to delayed outcome.

3.3. Agents and Baselines

RL agents. All six agents use a shared two-layer MLP (2×128 hidden units) and a fixed set of hyperparameters held identical across the four state sizes (Appendix B). A2C, PPO, and SAC use Stable-Baselines3, while DQN, Double-DQN, and A3C use matched custom PyTorch implementations. No per-dimension hyperparameter search is performed, so any performance difference across dimensions is attributable to the observation vector rather than to the training recipe.

Rule-based baselines span five management styles. The static-threshold baseline (`static_threshold`) applies one fixed shared threshold. The split-static baseline (`split_policy_static`) maintains separate fixed defaults per segment. The profit-oriented policy (`profit_oriented`) performs a weekly grid search maximising current-cohort expected profit without penalty terms. The risk-aware weekly policy (`risk_aware_weekly`) adjusts thresholds reactively to rolling default pressure and capital signals. The constraint-aware weekly policy (`constraint_aware_weekly`) runs a penalty-weighted weekly

Table 1: The six evaluation scenarios and their primary stress mechanism.

Scenario	Stress mechanism
base_market	Stable, the cleanest test of policy signal exploitation
drift	Gradual new-client score deterioration and rising defaults
adverse_stress	Abrupt shock: higher defaults, larger loans, capital pressure
class_imbalance_shift	Growing share of weaker new applicants, tests composition risk
noise	Stable mean with noisy weekly volumes and measurements
split_policy_dynamics	New clients deteriorate while repeat clients improve

grid search that explicitly internalises default-rate, approval-rate, capital-usage, and volatility violations.

3.4. Reward Function

With delayed reward enabled, the shaped weekly reward is:

$$\begin{aligned} r_t = \; & w_p\, \Pi_t^{\text{realized}} + w_n\, \text{NPV}_t^{\text{realized}} - w_s\, \hat{d}_t \cdot 100 \\ & - \lambda_d \big[\max\big(0,\, \bar{d}_t - 0.12\big)\big]^2 \cdot 1000 - \lambda_a\, |\alpha_t - 0.42| \cdot 1000 \\ & - \lambda_c \big[\max\big(0,\, c_t - 0.82\big)\big]^2 \cdot 1000 - \lambda_v \max\left(0,\, \frac{\sigma_{\Pi,t} - 8500}{8500}\right) \cdot 1000, \end{aligned} \tag{1}$$

where Π_t^{realized} and $\text{NPV}_t^{\text{realized}}$ are realized weekly profit and NPV (not current-cohort expectations), $\hat{d}_t$ is the expected default rate of the current cohort, $\bar{d}_t$ is the rolling realized default rate, α_t is the weekly approval rate, c_t is the projected capital usage ratio, and $\sigma_{\Pi,t}$ is realized profit volatility. The

four constants are calibration targets: the default-rate cap (0.12), the target approval rate (0.42), the capital-usage cap (0.82), and the profit-volatility scale (8500). Active weights: w_p=1.0, w_n=0.25, w_s=0.35, λ_d=9.0, λ_a=2.0, λ_c=3.5, λ_v=1.4.

The reward is multi-objective by design. Rankings by cumulative reward need not coincide with rankings by raw profit, because the reward encodes portfolio risk constraints rather than profit alone. Throughout this paper, "realized profit" is computed from realized interactive weekly profits plus terminal settlement, that is, realized final portfolio profit rather than a current-cohort expectation.

4. Experimental Design

The experiment is fully controlled: *only* observation dimensionality varies across the four runs. Everything else (the simulator, the reward, the eight random seeds, the 1,500 training episodes per seed, the evaluation protocol, the scenarios, the threshold grid, and the controller set) is held fixed. Each of the six agents is trained once per state dimension per seed (6 agents $\times$ 4 dimensions $\times$ 8 seeds = 192 training runs, each running the full 1,500-episode budget with the training scenario drawn from a pool of all six market scenarios), and evaluated over 6 scenarios $\times$ 8 seeds $\times$ 12 evaluation runs = 576 evaluation runs per controller per dimension. A validity layer programmatically verifies that the first 12 observation features are bit-for-bit identical across all four state sizes on a common reset and three fixed-action transitions.

In Table 2, the cumulative composition of the four observation layers is

Table 2: Cumulative observation layers. The first 12 features are invariant across all configurations and are verified programmatically before each experiment.

Size	Added information
12D	Compact dashboard: week progress, overall/segment approval rates, rolling realized default rate, expected default rate, realized profit, profit volatility, projected capital usage, outstanding ratio, both threshold levels.
20D	12D + current economic context: repeat segment share, expected profit/NPV per application, realized NPV, weekly reward, threshold gap, capital headroom, realized-vs-expected default gap.
30D	20D + lagged/rolling memory: second-lag approval/default/profit/capital, four-week rolling means and std of approval, default, and profit, lagged threshold changes.
50D	30D + segment-specific signals and stability descriptors: segment approval/ default/profit indicators, accepted-share ratios, rolling reward moments, cumulative reward/profit to date, capital-usage volatility, outstanding-ratio changes, threshold-gap history, application-volume ratio.

shown, where each layer adds features that a portfolio manager would realistically observe at weekly decision time with no access to future outcomes. Because each layer is a superset of the previous one, observation dimensionality is the only factor that changes across the four conditions.

4.1. Hypotheses

The three hypotheses follow from the research gap and from the literature reviewed in Section 2, and each is tested on paired run-level data in Section 5.4. The first hypothesis concerns state representation. Because feature enhancement and richer inputs are repeatedly reported to help credit models up to a point (Yang et al., 2024), we expect added observation features to improve the RL frontier but only over an intermediate range. The second hypothesis concerns the RL advantage. The credit-scoring benchmarking literature repeatedly finds that more complex models do not reliably dominate simpler ones (Lessmann et al., 2015; Gunnarsson et al., 2021), and the direct precursor compared RL against only a single static baseline (Herasymovych et al., 2019; Kiatsupaibul et al., 2024), so we expect the universal-dominance claim to fail against strong rule-based policies. The third hypothesis concerns delayed evaluation. Because outcomes are delayed and terminal settlement is applied, as in the delayed-reward credit-control setting of Herasymovych et al. (2019), early-period performance may misrepresent final portfolio quality.

H1 (State representation).

Richer state representations improve the best RL frontier from 12D to 30D, while the move from 30D to 50D does not necessarily create additional improvement.

H2 (RL vs. rule-based).

RL improves over rule-based policies only in some scenarios and metrics, and the universal claim that RL robustly dominates strong rule-based baselines is expected to be *rejected*.

H3 (Delayed evaluation).

Early-period performance is an unreliable proxy for final realized portfolio quality when outcomes are delayed and terminal settlement is applied. The "locally worse, globally better" pattern is expected to hold.

5. Results

5.1. Best Overall and Best RL Controller by Dimension

In Table 3, the strongest controller is reported when baselines and RL agents are pooled under the repository's official ranking rule (highest cumulative reward, then highest profit, then lowest default rate), together with the strongest RL controller in each dimension (full-profile run, 8 seeds $\times$ 1,500 episodes per seed, with data and configuration in the public repository). Unlike under the short 50-episode budget, the best RL agent is not the same algorithm at every dimension: SAC leads at 12D and 50D, A2C at 20D, and DQN at 30D. The best RL profit still peaks at 12D and dips at 30D before partially recovering at 50D, but the gap to the leading baseline is now small at every dimension.

By cumulative reward, the primary ranking metric, the constraint-aware weekly policy is the best overall controller in every dimension (reward 3,666,052). On raw realized profit it is statistically tied with the profit-oriented policy, which is marginally higher (3,803,168 vs. 3,802,547). The two differ by less than 0.02% and the difference is not economically meaningful. The flat baseline row confirms that the experiment isolates the effect of observation size on RL quality, because the leading rule-based policy is independent of the RL state vector by construction. With the 1,500-episode budget, the profit

Table 3: Best overall controller and best RL controller by state dimension (full-profile run, 1,500 training episodes per seed). The baseline line is flat because the leading baseline does not depend on the RL observation vector. "Profit" is realized final portfolio profit (Section 3.4); "Reward" is cumulative shaped reward.

Dim	Controller	Type	Profit	Reward	Default rate
Best overall (pooled, by cumulative reward)					
12	Constraint-aware weekly	baseline	3,802,547	3,666,052	0.0760
20	Constraint-aware weekly	baseline	3,802,547	3,666,052	0.0760
30	Constraint-aware weekly	baseline	3,802,547	3,666,052	0.0760
50	Constraint-aware weekly	baseline	3,802,547	3,666,052	0.0760
Best RL agent					
12	SAC	RL	3,785,010	3,631,406	0.0768
20	A2C	RL	3,761,562	3,624,570	0.0768
30	DQN	RL	3,706,594	3,633,945	0.0749
50	SAC	RL	3,724,499	3,605,339	0.0762

gap between the best RL agent and the best overall controller shrinks to −17,537 (−0.46%) at 12D, −40,985 (−1.08%) at 20D, −95,953 (−2.52%) at 30D, and −78,049 (−2.05%) at 50D, and the corresponding reward gaps are −34,646 (−0.95%), −41,482 (−1.13%), −32,108 (−0.88%), and −60,713 (−1.66%). Every gap is under 2.6% of the baseline value, compared with −9.4% for the best RL agent's 12D peak under the short 50-episode budget, so the dominant effect of the larger training budget is to close most of the RL-to-baseline gap rather than to change which controller wins. Default rate for the best RL agent is 7.68% at 12D, 7.68% at 20D, 7.49% at 30D, and 7.62% at 50D, again lowest at 30D, which foreshadows the risk-side pattern

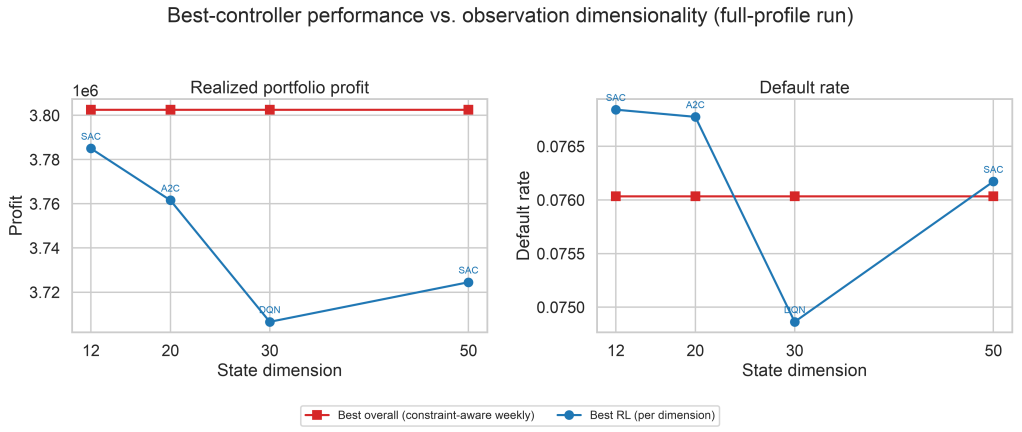


Figure 3: Best-controller performance versus observation dimensionality (full-profile run, 8 seeds, 1,500 training episodes per seed). *Left*: realized portfolio profit. *Right*: default rate. The best overall controller (the constraint-aware weekly policy) is independent of the RL state vector and is therefore flat across dimensions, while the best RL agent (labelled per point: SAC at 12D and 50D, A2C at 20D, DQN at 30D) sits close to the baseline on profit at every dimension and crosses below the baseline on default rate at 30D and 50D.

developed in Section 5.3.

Figure 3 shows the same comparison as a function of observation dimensionality. The figure shows that the best RL agent's profit is close to the leading baseline at every dimension, with a shallow dip at 30D, while on default rate the best RL agent moves from above the baseline line at 12D and 20D to below it at 30D and 50D – richer state does not change who wins the reward ranking, but it does change whether RL's default rate is competitive with the profit-seeking baselines.

5.2. Agent Comparison

In Table 4, the mean performance of each agent across all four dimensions is summarised alongside the three principal rule-based baselines (full-profile

Table 4: Agent and baseline summary (mean across all four dimensions, full-profile run, 1,500 training episodes per seed). "Approval" denotes mean weekly approval rate. Baseline rows do not depend on state dimension, so their per-dimension values are identical and the mean equals any single dimension's value.

Controller	Type	Mean profit	Default rate	Approval
Profit-oriented	baseline	3,803,168	7.61%	94.91%
Constraint-aware weekly	baseline	3,802,547	7.60%	94.84%
Static-threshold	baseline	3,083,370	6.62%	69.99%
SAC	RL	3,700,732	7.64%	92.58%
A2C	RL	3,607,319	7.59%	90.12%
A3C	RL	3,472,312	7.17%	83.28%
DQN	RL	3,393,809	7.36%	82.68%
Double-DQN	RL	3,390,977	7.24%	81.97%
PPO	RL	2,617,957	6.21%	57.16%

run, 1,500 training episodes per seed). With this larger budget the RL ranking is very different from a short-budget run: SAC leads all RL agents on mean profit and comes within 2.7% of the best baseline's mean profit, PPO remains the weakest and most degenerate agent, and the DQN family is no longer the strongest RL agent on average even though it supplies the canonical reference agent for the hypothesis tests in Section 5.4.

SAC leads all RL agents on mean profit (3,700,732), only 2.7% below the leading baseline's mean profit (3,803,168), a striking change from the -9.4% gap of the best RL agent under a short 50-episode budget. A2C is a close second (3,607,319). DQN and Double-DQN, the two off-policy value-based agents, are now mid-table on mean profit (3,393,809 and 3,390,977) despite

DQN winning outright at 30D (Table 3), which shows that a 30D-specific advantage does not carry over to the other three dimensions for this agent family. PPO remains the weakest and most degenerate RL agent: its profit, default rate, and approval rate are numerically identical across 12D, 20D, 30D, and 50D (2,617,957 profit at every dimension), which means PPO converges to the same policy regardless of the additional observation features even under the larger training budget, so those features remain unused rather than uninformative. Between the two extremes, mean default rate tracks mean approval rate across agents (higher-approval agents accept more marginal applicants and so post higher default rates), which is consistent with a credit-quality trade-off along the approval axis rather than a free improvement from any single agent.

5.3. Scenario-Level Winners

In Table 5, the reward-based, profit-based, and lowest-default-rate winner for each scenario at the 50D configuration is reported. No single metric identifies one controller as best, because the reward, profit, and default-rate winners differ across scenarios.

RL wins zero of six scenarios by reward at every dimension, and zero of six by profit at 20D, 30D, and 50D. At 12D, however, SAC wins raw profit in two of six scenarios (class_imbalance_shift, by 4,248, and drift, by 802, both narrow margins against the profit-oriented policy), the only dimension at which an RL agent tops the profit column for any scenario. The profit-oriented policy wins raw profit in the remaining scenarios at every dimension, which is why the headline "best overall" ranking, based on cumulative reward, and the profit ranking need not coincide.

Table 5: Scenario-level winners at 50D state (full-profile run, 1,500 training episodes per seed; the reward and default columns hold at every dimension, the profit column holds at 20D/30D/50D, see text for the 12D exception). *Reward winner*: highest cumulative shaped reward. *Profit winner*: highest realized profit. *Default winner*: lowest default rate. The risk-aware weekly policy is the single lowest-default controller in every scenario at every dimension.

Scenario	Reward winner	Profit winner	Default winner
adverse_stress	Static-threshold	Constraint-aware weekly	Risk-aware weekly
base_market	Profit-oriented	Profit-oriented	Risk-aware weekly
class_imbalance_shift	Constraint-aware weekly	Profit-oriented	Risk-aware weekly
drift	Profit-oriented	Profit-oriented	Risk-aware weekly
noise	Profit-oriented	Profit-oriented	Risk-aware weekly
split_policy_dynamics	Constraint-aware weekly	Profit-oriented	Risk-aware weekly

The default-rate advantage of the best RL agent over the two profit-seeking baselines (profit-oriented and constraint-aware weekly) is strongly dimension-dependent. Counting per-scenario wins against *both* profit-seeking baselines simultaneously, the best RL agent undercuts them in 0 of 6 scenarios at 12D (SAC), 1 of 6 at 20D (A2C, in the noise scenario only), and 5 of 6 at both 30D (DQN) and 50D (SAC), losing only in adverse_stress at every dimension. So the "richer-state RL undercuts the profit-seeking baselines on default rate" pattern is not a property of RL in general: it appears once the observation is rich enough (30D and above) and is essentially absent at 12D and weak at 20D. The risk-aware weekly policy remains the single lowest-default controller in every scenario at every dimension (Table 5). This result has direct practical significance: a portfolio manager willing to accept a

small profit reduction in exchange for capital conservation may prefer a 30D-or-larger RL policy over the profit-seeking baselines, but should not expect the same benefit from a 12D or 20D RL deployment.

5.4. Statistical Hypothesis Tests

In Table 6, the paired statistical comparisons for the three hypotheses of Section 4.1 are summarised, all computed on the full-profile run (1,500 training episodes per seed), with the paired run-level and weekly-trajectory data in the public repository. The tests confirm that 30D significantly beats both 12D and 50D, that RL (represented by the reference DQN agent) is still significantly weaker than the profit-oriented baseline on reward at 30D and 50D though the 30D gap is now small, and that the RL deficit under stress reverses well before the end of the horizon. The dimensionality hypotheses are tested on vanilla **DQN**, used here as the canonical discrete-action reference agent for the controlled state-size comparison; DQN wins outright at 30D (Table 3) but is not the best mean RL agent overall (Table 4, where SAC leads), so DQN is used consistently as the fixed reference rather than as a claim that it is the strongest agent.

H1 (richer state improves RL up to an intermediate representation) is *fully supported* on the reference agent, a reversal from the short-budget run. The 30D to 12D reward difference is now positive and highly significant ($p < 10^{-3}$, mean +138,516, $n = 1{,}296$ paired runs), so 30D reliably beats 12D under the larger training budget. The 30D to 50D comparison remains highly significant ($p < 10^{-3}$, mean +662,993, an even larger margin than before), which confirms that 50D reliably underperforms 30D. A supplementary check against the 20D best-RL frontier (Double-DQN) tells a more nuanced story:

Table 6: Paired statistical checks (full-profile run, 1,500 training episodes per seed). The H3 rows use the dim-30 weekly trajectories in the adverse_stress scenario; week 25 is roughly the midpoint of the 52-week interactive horizon (interactive weeks are indexed 0–51), not the final week. *Mean diff.*: left minus right controller, where positive values favour the left controller on reward/profit and negative values favour it on default rate. CI: 95% percentile bootstrap.

Hyp.	Paired comparison	Metric	Mean diff.	95% CI	p
H1	30D DQN − 12D DQN	Reward	+138,516	[+124,800, +151,557]	$<10^{-3}$
H1	30D DQN − 50D DQN	Reward	+662,993	[+639,441, +685,224]	$<10^{-3}$
H2	30D DQN − Profit-oriented	Reward	−25,134	[−34,282, −15,755]	$<10^{-3}$
H2	50D DQN − Profit-oriented	Reward	−688,127	[−712,048, −665,442]	$<10^{-3}$
H3	DQN − Static-threshold (week 6, early)	Profit	−853	[−2,568, +954]	0.345
H3	DQN − Static-threshold (week 25, mid-horizon)	Profit	+152,941	[+140,408, +166,059]	$<10^{-3}$

30D DQN beats 20D Double-DQN on reward (+284,728, $p < 10^{-3}$) but has a significantly *higher* default rate by 0.76 percentage points ($p < 10^{-3}$), so the 30D advantage over 20D is a reward-side, not a risk-side, result on this specific pairing.

H2 (RL does not universally dominate) remains confirmed, but the margin now depends heavily on dimension. The 30D DQN agent is significantly weaker than the profit-oriented policy on reward, but by only −25,134 ($p < 10^{-3}$, a 0.7% gap), down from −368,145 under the short-budget run. The 50D DQN agent, in contrast, trails by −688,127 ($p < 10^{-3}$), a large gap that reflects DQN's comparatively weak mean performance at 50D (Table 4)

rather than a property of RL at 50D in general, since SAC is the best RL agent at 50D and comes much closer to the baseline (Table 3). On the risk side, 30D DQN has a significantly *lower* default rate than the profit-oriented policy (-0.083 percentage points, $p < 10^{-3}$), consistent with the scenario-level pattern in Section 5.3.

H3 (delayed evaluation changes rankings, "locally worse globally better") is *supported*, and more strongly than under the short-budget run. The H3 checkpoints are computed directly on the full-profile dim-30 run's weekly trajectories in the adverse_stress scenario ($n = 216$ paired observations at each checkpoint), not on a separate development profile. At the early checkpoint (interactive week 6) DQN is statistically indistinguishable from the static-threshold baseline (profit gap -853, $p = 0.345$), essentially tied rather than trailing. By week 25, roughly the midpoint of the 52-week interactive horizon, DQN has opened a large and highly significant advantage over the static baseline (profit gap $+152{,}941$, $p < 10^{-3}$). The pattern is therefore "locally tied, globally better" rather than the persistent deficit found under the short-budget run: the RL agent's advantage under stress accumulates over the horizon as delayed outcomes settle, and it is already decisive well before the horizon ends, which argues against relying on early-horizon snapshots to judge RL policies under delayed-outcome evaluation.

6. Discussion

6.1. Why Rule-Based Baselines Still Win

The leading baselines (the constraint-aware weekly and profit-oriented policies) perform a weekly grid search over the same 121 joint threshold

actions as the RL agents. Their advantage is structural: they evaluate each action directly on the current week's cohort economics, they see the realized profit and default signals immediately, and they set thresholds according to an explicit objective, so they need no training at all. Raising the training budget from 50 to 1,500 episodes per seed closes most, but not all, of the gap: the best RL agent's profit shortfall shrinks from -9.4% to between -0.5% and -2.5% depending on dimension (Table 3), and the reference DQN agent's reward deficit against the profit-oriented policy at 30D falls from $-368{,}145$ to $-25{,}134$ (Section 5.4). This shows that training budget, not some fundamental ceiling on RL, was the dominant factor behind the earlier short-budget gap. What remains after $30\times$ more training is a much smaller, but still statistically significant, residual gap on every scenario and dimension tested, consistent with a genuine structural advantage for direct weekly evaluation over a learned value function under delayed, noisy rewards, rather than an artifact of undertraining.

6.2. The Conditional Value of RL

Whether RL agents undercut the two profit-seeking baselines on default rate is strongly dimension-dependent, not a general property of RL. Counting scenarios in which the best RL agent beats both the profit-oriented and constraint-aware weekly policies on default rate simultaneously, the pattern is 0 of 6 scenarios at 12D, 1 of 6 at 20D, and 5 of 6 at both 30D and 50D (Section 5.3); the risk-aware weekly policy remains the lowest-default controller overall in every case (Table 5). A portfolio manager who values capital conservation should therefore not expect a risk-side benefit from a 12D or 20D RL deployment, but may accept a modest profit reduction in exchange

for a real risk-side improvement once the observation is rich enough (30D or larger). This portfolio-quality advantage is not visible in the headline reward metric, and it is conditional on both the training budget and the observation dimensionality, not on RL as a method in general.

6.3. Dimensionality Saturation

The 30D state adds lagged and rolling history (second-lag approval, default, and profit values, four-week rolling means and standard deviations, and lagged threshold changes). This temporal context is what a controller needs under slow-moving drift or noisy volume signals, namely the ability to compare this week with the recent past. The 50D state adds richer segmentation and stability descriptors, but those extra features require more optimization work and offer diminishing marginal information at the current training budget. With the larger, 1,500-episode budget, 30D is no longer only a risk-side compromise: it is now the reward-side optimum for the reference DQN agent as well, since 30D significantly beats *both* 12D and 50D on cumulative reward (Section 5.4), a stronger result than the earlier short-budget run, in which 30D beat only 50D and was statistically indistinguishable from 12D. The result is that 30D is the most balanced point on the *risk-and-information* axis: it delivers more temporal memory than 12D while avoiding the regression that 50D introduces for the DQN family, and it is the only dimension at which the reference agent's own reward peaks. The best-per-dimension profit picture is more mixed, since SAC (not DQN) is the strongest agent at 12D and 50D and edges out DQN's 30D profit in absolute terms (Table 3), so the case for 30D as a default rests on the reference-agent reward result and the risk-side benefit developed above, not on every algorithm peaking there.

6.4. Episode-Static RL Behavior

Under the short 50-episode budget, every best-per-dimension RL agent had threshold volatility of 0.00, and the finding was that RL policies are uniformly episode-static. Under the larger 1,500-episode budget this is no longer true: three of the four best-per-dimension agents show real within-episode threshold adjustment (SAC at 12D, volatility 2.10; A2C at 20D, 1.27; SAC at 50D, 4.20), and only DQN at 30D remains exactly static (0.00). Looking at all six agents across all four dimensions, the pattern is algorithm-specific rather than dimension-specific: DQN and Double-DQN (the two off-policy, greedy value-based agents) stay at 0.00 threshold volatility at 12D, 20D, and 30D and only begin adjusting thresholds within the episode at 50D (volatility 4.94 and 4.98 respectively); PPO remains at 0.00 at every dimension, consistent with its degenerate, dimension-invariant policy (Section 5.2); while SAC and the two actor-critic agents (A2C, A3C) show nonzero within-episode adjustment at every dimension, generally increasing with state size for SAC (2.10 at 12D up to 4.20 at 50D). Episode-static behavior is therefore a property of the DQN family's greedy, off-policy exploitation at low-to-moderate dimensionality, not a general RL phenomenon: entropy-regularized and on-policy actor-critic agents adapt within the episode throughout, and even the DQN family does so once the state is large enough. The implication for practitioners is that the choice of algorithm, not just training budget or state size, determines whether a controller performs genuine week-to-week adaptation or effectively episode-level threshold selection.

6.5. Practical Recommendations

For a lending business the results translate into concrete guidance.

- Training budget matters more than the choice between RL and rule-based search: moving from 50 to 1,500 training episodes per seed cut the best RL agent's profit shortfall from -9.4% to under -2.6% at every dimension, so an institution considering RL should budget for substantial training before judging the approach on results from a short pilot run.
- The weekly grid-search controller (the constraint-aware or profit-oriented weekly policy) still wins the primary reward ranking in every scenario and dimension and needs no training, so it remains the safe default when training resources or validation time are limited.
- Adopt an RL controller, specifically at 30D or 50D, when the institution values default-rate reduction and can accept a small profit trade-off, because only at these dimensions does the best RL agent undercut both profit-seeking baselines on default rate in a majority of scenarios (5 of 6); the same benefit should not be expected at 12D (0 of 6) or 20D (1 of 6).
- Use a 30D state as a sensible default when RL is deployed: it is the only dimension at which the reference DQN agent's own reward peaks, it supports the risk-side benefit above, and it avoids the regression that 50D introduces for the DQN family. If SAC or an actor-critic agent is preferred instead of DQN, 12D or 50D are also strong candidates, since the best per-dimension agent varies by algorithm.

6.6. Limitations

First, all results derive from a synthetic simulator, so calibration to any specific production portfolio would shift the absolute numbers and potentially

alter some rankings. Second, although the 1,500-episode-per-seed training budget is 30× larger than the short-budget run and closes most of the RL-to-baseline gap, it is not a proof of full convergence; a still larger budget could narrow the residual gap further, particularly for the DQN family at 50D. Third, controller rankings are sensitive to the reward weights in Equation (1), and a reward sensitivity analysis is left for future work. Fourth, bootstrap confidence intervals quantify simulation uncertainty, not real-world external validity. Fifth, the H3 delayed-evaluation check is reported for a single scenario (adverse_stress) and a single dimension (30D); whether the same "locally tied, globally better" pattern holds in other scenarios and dimensions is left for future work.

These results are bounded by the synthetic setting. The default and recovery parameters are calibrated to aggregate statistics rather than fitted to a single proprietary book, so the absolute profit and default levels should be read as order-of-magnitude rather than exact. Even at 1,500 training episodes per seed, a still larger budget could narrow the residual RL-to-baseline gap further, so the comparison in this paper should be read as "RL under a substantial, but not unlimited, training budget" rather than as an asymptotic comparison. The real-data comparison validates the score-to-default relationship and the parameter ranges, but it does not validate the weekly threshold dynamics directly, which would require a lender to deploy and log a controlled threshold policy.

7. Threats to validity

The conclusions of this study rest on a synthetic simulator, so their external validity depends on how faithfully the simulator reproduces the dynamics of real unsecured consumer lending. This section states the main threats, compares the simulator against published real-world evidence, and explains why an agent-based simulator remains the appropriate tool under these threats.

7.1. Construct validity and simulator fidelity

The portfolio is generated from parametric assumptions rather than observed loans, which introduces three concrete risks. First, the logistic default model in Appendix A imposes a smooth monotone link between score and default probability, whereas real default curves can be locally non-monotone in thin score regions. Second, the weekly event schedule collapses repayment, default, and recovery into a small set of timed cashflows, which understates the heterogeneity of real repayment paths. Third, the scenario set spans six stylised regimes, so it cannot represent every macro-financial path a real book would face. We mitigate these risks by calibrating every primary parameter to published statistics and by comparing the score-to-default relationship against a large real loan book, as described next.

7.2. Comparison with published evidence and a real loan book

Real consumer credit portfolios show a wide spread of default rates and strong class imbalance (Brown & Mues, 2012), which the scenario set is designed to span. Table 7 places each primary simulator parameter next to the corresponding real-world range and its source. In Table 7, the portfolio

default rate, the loan tenor, the effective interest, and the recovery assumption all fall inside the ranges reported for unsecured retail lending, and the acceptance-threshold mechanism reproduces the setup of Herasymovych et al. (2019), whose optimal acceptance threshold on a real online consumer-loan book sits in the same mid-score region used here. Figure 4 overlays the simulator default-probability curves on the empirical default rate by within-book credit-score percentile estimated from the Lending Club loan book (Xia et al., 2017). In Figure 4, plotting against the within-book percentile aligns the two score domains, and both relationships fall monotonically while the higher Lending Club levels reflect its higher-rate and longer-tenor borrower mix.

Table 7: Calibration of the simulator against published evidence. In Table 7, each primary parameter sits inside a real-world range, and the sources are aggregate statistics and peer-reviewed studies rather than a single proprietary book.

Parameter	Simulator	Real-world range (source)
Portfolio default rate	4–14% across scenarios	around 4–5% in normal conditions and up to 12.9% under stress for unsecured consumer loans (Bank of Russia, 2025) 20.0% on Lending Club resolved loans (own computation; Xia et al., 2017)
Acceptance mechanism	weekly score threshold, grid 35–85	optimal threshold in the mid-score region on a real online book (Herasymovych et al., 2019)
Loan tenor	4–12 months	short-term consumer and online installment products (Herasymovych et al., 2019) Lending Club is longer, with 76% at 36 months and 24% at 60 months
Loan amount	0.3–9.0 k currency units	small-ticket loans typical of short-term online consumer lending (Herasymovych et al., 2019) Lending Club median is larger, at 12,000 USD
Effective interest	11–15% annualised	comparable to the Lending Club median of 12.7%
Recovery on default	26–34% (LGD $\approx$ 66–74%)	recovery distributions for unsecured consumer loans (Loterman et al., 2012; Qi & Zhao, 2011)

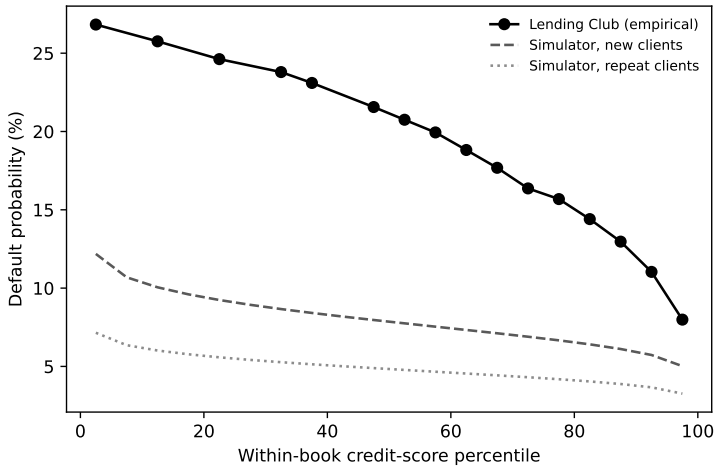


Figure 4: Default probability by within-book credit-score percentile, simulator curves versus the Lending Club loan book. In Figure 4, the within-book percentile axis aligns the two score domains, and the higher Lending Club levels reflect its higher-rate and longer-tenor borrower mix.

7.3. Why an agent-based simulator

Loan-level data that jointly record the score, the funding decision, the realised outcome, and the calendar timing across several market regimes are rarely public, because lender underwriting models and outcome histories are proprietary assets (Di Maggio et al., 2024). In the absence of such data, an agent-based simulator acts as a controlled laboratory that varies one factor at a time, here the observation dimension, while holding the protocol fixed, which is the comparison that observational data cannot deliver (Axtell & Farmer, 2025). We therefore read the simulator as a tool for studying the structure of the control problem under a fixed training budget, not as a forecast of any single lender's profit.

8. Conclusion

This paper evaluates six reinforcement learning algorithms against five rule-based baselines across six market scenarios and four observation-space sizes (12D, 20D, 30D, 50D) in a controlled credit portfolio simulation, at a training budget of 1,500 episodes per seed. The main empirical findings are:

(1) The constraint-aware weekly policy remains the best overall controller in every dimension by cumulative reward (reward 3,666,052). On raw realized profit it is statistically tied with the profit-oriented policy (3,802,547 vs. 3,803,168, the latter marginally higher). RL wins zero of six scenarios by reward across all dimensions, and zero of six by profit at 20D, 30D, and 50D, but two of six at 12D (SAC beats the profit-oriented policy in class_imbalance_shift and drift, by narrow margins).

(2) The best RL agent is not a single algorithm: SAC leads at 12D (profit 3,785,010) and 50D (3,724,499), A2C at 20D (3,761,562), and DQN at 30D (3,706,594). The profit gap to the best overall controller shrinks to between -0.46% (12D) and -2.52% (30D) at every dimension, down from -9.4% for the best RL agent (Double-DQN, 12D) under a $30\times$ smaller training budget.

(3) The 30D to 12D reward improvement for the reference DQN agent is now positive *and* statistically significant ($+138{,}516$, $p < 10^{-3}$), a reversal from the short-budget run, and 30D continues to significantly outperform 50D ($+662{,}993$, $p < 10^{-3}$), so 30D is the reward-side optimum for this reference agent, not only a risk-side compromise.

(4) The best RL agent undercuts the two profit-seeking baselines on default rate in a majority of scenarios (5 of 6) at 30D and 50D, but this risk-

side advantage is weak at 20D (1 of 6) and absent at 12D (0 of 6); the risk-aware weekly policy stays the lowest-default controller overall at every dimension. The conditional risk-side advantage of RL is therefore dimension-dependent, not universal.

(5) A delayed-evaluation check in the adverse-stress scenario shows DQN statistically tied with the static-threshold baseline early in the horizon (week 6, $p = 0.345$) and significantly ahead by the horizon midpoint (week 25, +152,941, $p < 10^{-3}$), a "locally tied, globally better" pattern, the opposite of the persistent early-and-late deficit found under the short-budget run.

(6) Episode-static behavior, in which threshold volatility is 0.00 throughout the episode, is no longer universal: it is specific to the DQN family at 12–30D (which only begins adjusting thresholds within the episode at 50D) and to PPO at every dimension, while SAC and the actor-critic agents (A2C, A3C) adjust thresholds within the episode at every dimension tested.

The practical implication has shifted from the short-budget run: training budget, not a fundamental RL-versus-rule-based ceiling, was the dominant factor behind the earlier large gap. Well-designed rule-based weekly search still wins the primary reward ranking in every scenario and dimension tested and needs no training, so it remains the safe default when training resources are limited, but a sufficiently trained RL agent is now a credible alternative on profit, and specifically at 30D or 50D it also delivers a real risk-side (lower default rate) benefit in most scenarios. The 30D observation space is the strongest default for an RL deployment: it is the only dimension at which

the reference DQN agent's own reward peaks, and it supports the risk-side advantage developed above. Future work should investigate whether an even larger training budget closes the remaining -0.46% to -2.52% profit gap entirely (particularly for the DQN family at 50D), reward sensitivity to penalty weights, more adaptive recurrent or attention-based architectures, whether the "locally tied, globally better" pattern generalizes beyond the adverse-stress/30D pairing tested here, and validation on real or semi-synthetic loan panels.

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Appendix A. Simulator details

This appendix specifies the data-generating process so that the experiments can be reproduced from the public repository. All values are the calibrated full-profile settings.

Applicant generation.. Each interactive week the simulator draws a base application volume around 600 applications, scales it by a seasonal factor and multiplicative volume noise, and splits it into new and repeat counts through two Poisson draws. The repeat share follows a scenario base level modulated by an annual seasonal term and clamped to $[0.15, 0.75]$. Each applicant in segment s receives a credit score drawn from a Gaussian $\mathcal{N}(\mu_s, \sigma_s)$ clipped to $[0, 100]$, with $\mu_{\text{new}} = 59$, $\sigma_{\text{new}} = 13$ and $\mu_{\text{repeat}} = 69$, $\sigma_{\text{repeat}} = 10$ in the base market.

Default and recovery.. An applicant with score x in segment s defaults with probability

$$p_s(x) = \sigma\big(\beta_s - \kappa_s \cdot \tfrac{x-50}{10}\big)\,, \tag{A.1}$$

clipped to $[0.01, 0.98]$, where σ is the logistic function and $\beta_{\text{new}} = -2.3$, $\kappa_{\text{new}} = 0.18$, $\beta_{\text{repeat}} = -2.6$, $\kappa_{\text{repeat}} = 0.20$ in the base market. Scenario stress shifts β_s gradually over the horizon and an abrupt shock adds to β_s from the shock week onward (Table A.8). These settings target portfolio default rates of 4–6% in the base market and 9–14% under adverse stress. A defaulted loan recovers with probability $r_{\text{new}} = 0.26$ or $r_{\text{repeat}} = 0.34$, and on recovery the lender receives the principal plus a fraction φ_s of the scheduled interest, with $\varphi_{\text{new}} = 0.65$ and $\varphi_{\text{repeat}} = 0.72$.

Loan terms and cashflows.. The principal is lognormal with mean $A_{\text{new}} = 1650$ or $A_{\text{repeat}} = 2150$ currency units and log-scale 0.28 (new) or 0.25 (repeat), clipped to $[300, 9000]$ and scaled by the scenario loan-amount multiplier. Each loan receives an integer term d in months (new: 5–12, repeat: 4–10) and a scheduled interest income of principal $\times$ $(\rho_s\, d/12 + o_s)$, with annual fee rate $\rho_{\text{new}} = 0.16$, $\rho_{\text{repeat}} = 0.13$ and origination $o_{\text{new}} = 0.035$, $o_{\text{repeat}} = 0.03$. Outcomes enter a forward event queue: a repaid loan returns the interest at week issue + d, a defaulted loan returns $-$principal at week issue + d, and a recovered loan adds the recovery value a further 1–4 weeks later. The term index d therefore sets both the interest horizon $d/12$ years and the event delay d weeks. Each cashflow is discounted to net present value at a weekly rate of $0.14/52$, that is an annual discount rate of 14%.

Episode structure and capital.. Each episode runs an 8-week warm-up at the default thresholds (60 for new, 50 for repeat), then a 52-week interactive horizon, then a terminal settlement that resolves all pending events so that loans originated near the horizon are not penalised. Outstanding principal

plus the principal accepted in the current week is tracked against a capital limit of 3,000,000 to form the capital-usage ratio used in the reward.

Table A.8: Per-scenario stress parameters. In Table A.8, the score and default shifts are total drifts applied gradually over the horizon, and the shock column gives the magnitude added to the default intercept and the week it begins.

Scenario	Repeat share	Score shift (new/rep)	Default shift (new/rep)	Score noise	Loan mult.	Shock (week)
base_market	0.35	0.00 / 0.00	0.00 / 0.00	1.00	1.00	none
drift	0.34	−0.18 / −0.07	0.18 / 0.08	1.00	1.00	0.08 (13)
adverse_stress	0.33	−0.10 / −0.10	0.30 / 0.22	1.15	1.15	0.22 (10)
class_imbalance_shift	0.28	−0.08 / 0.03	0.12 / −0.02	1.00	0.95	0.12 (16)
noise	0.35	0.00 / 0.00	0.04 / 0.04	1.45	1.00	none
split_policy_dynamics	0.38	−0.20 / 0.10	0.22 / −0.10	1.00	1.00	0.06 (14)

Appendix B. Training hyperparameters

In Table B.9, the shared training configuration for the full-profile run is listed. The settings are held identical across all four state sizes, so any performance difference between dimensions is attributable to the observation vector rather than to the training recipe.

Table B.9: Shared training settings (full-profile run), held identical across all four state sizes. In Table B.9, these settings are fixed so that any difference across dimensions reflects the observation vector rather than the training recipe.

Parameter	Value	Notes
Network architecture	MLP, 2×128 hidden units	Shared across all agents
Discount factor γ	0.98	Standard
Learning rate	SB3 default (varies by agent)	Not tuned
Training episodes/seed	1,500	Full profile
Seeds	11, 23, 37, 51, 67, 79, 97, 113	8 seeds, fixed across dims
Evaluation runs	12 per seed per scenario	Same protocol across dims
Threshold grid	$[35, 85]$ in steps of 5	11 actions per segment
Warm-up weeks	8	Pre-fills loan book
Interactive horizon	52 weeks	Full profile
Applications per week	600	Before scale factors
Bootstrap resamples	1000	95% CI

Table B.10: Per-agent reinforcement-learning hyperparameters (full-profile run), held identical across all four state sizes. In Table B.10, A2C, PPO, and SAC use Stable-Baselines3 (Raffin et al., 2021), while DQN, Double-DQN, and A3C use matched custom PyTorch implementations.

Agent	Implementation	Key hyperparameters
DQN	Custom	lr 10^{-3}, batch 64, replay 2×10^4, learning starts 300, train freq 1, gradient steps 1, target update 200, ε $1.0 \rightarrow 0.05$ over 65% of training
Double-DQN	Custom	As DQN, with a double-Q target
PPO	SB3	lr 3×10^{-4}, rollout $n = 256$, batch 64, GAE λ 0.95, entropy coef 0.01, clip range 0.20
A2C	SB3	lr 7×10^{-4}, rollout $n = 256$, GAE λ 0.95, entropy coef 0.01
A3C	Custom	lr 5×10^{-4}, entropy coef 0.01, value coef 0.5, full-episode rollouts (single worker)
SAC	SB3	lr 3×10^{-4}, buffer 5×10^4, learning starts 500, batch 128, τ 0.02, train freq 1, gradient steps 1

CRediT authorship contribution statement

Evgeny Baulin: Software, Investigation, Writing – original draft. **P. P. Lukianchenko:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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